

ECEN 607: Advanced Analog Circuit Tech

Lab 7: Current Reference and Bandgap Circuits

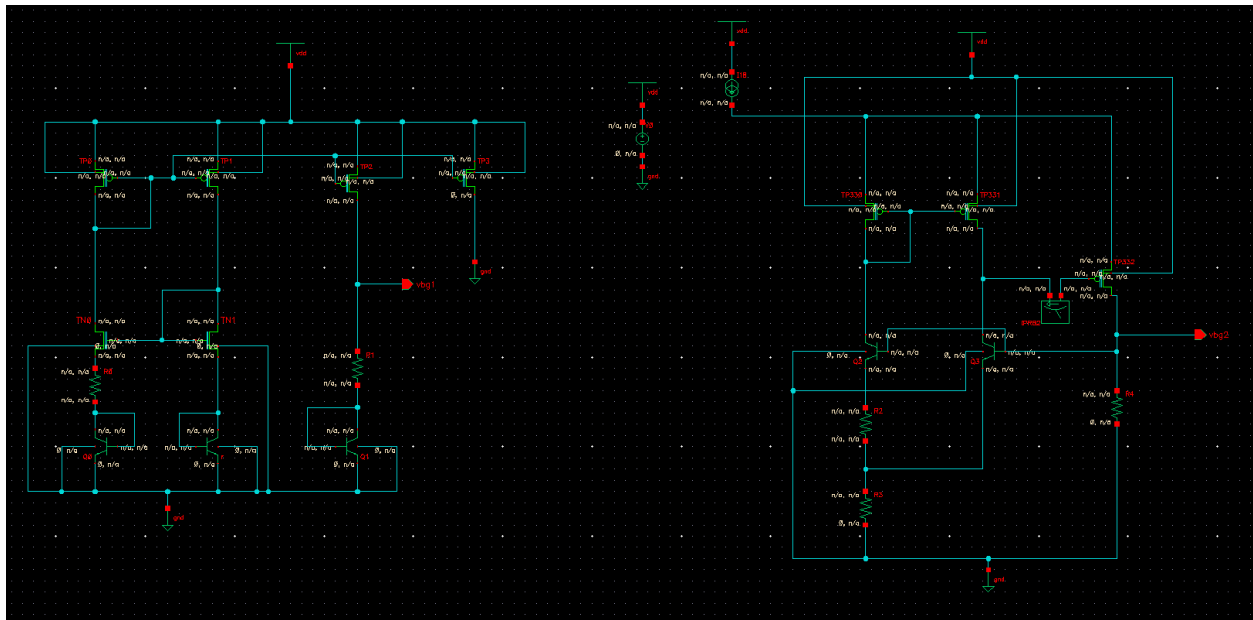
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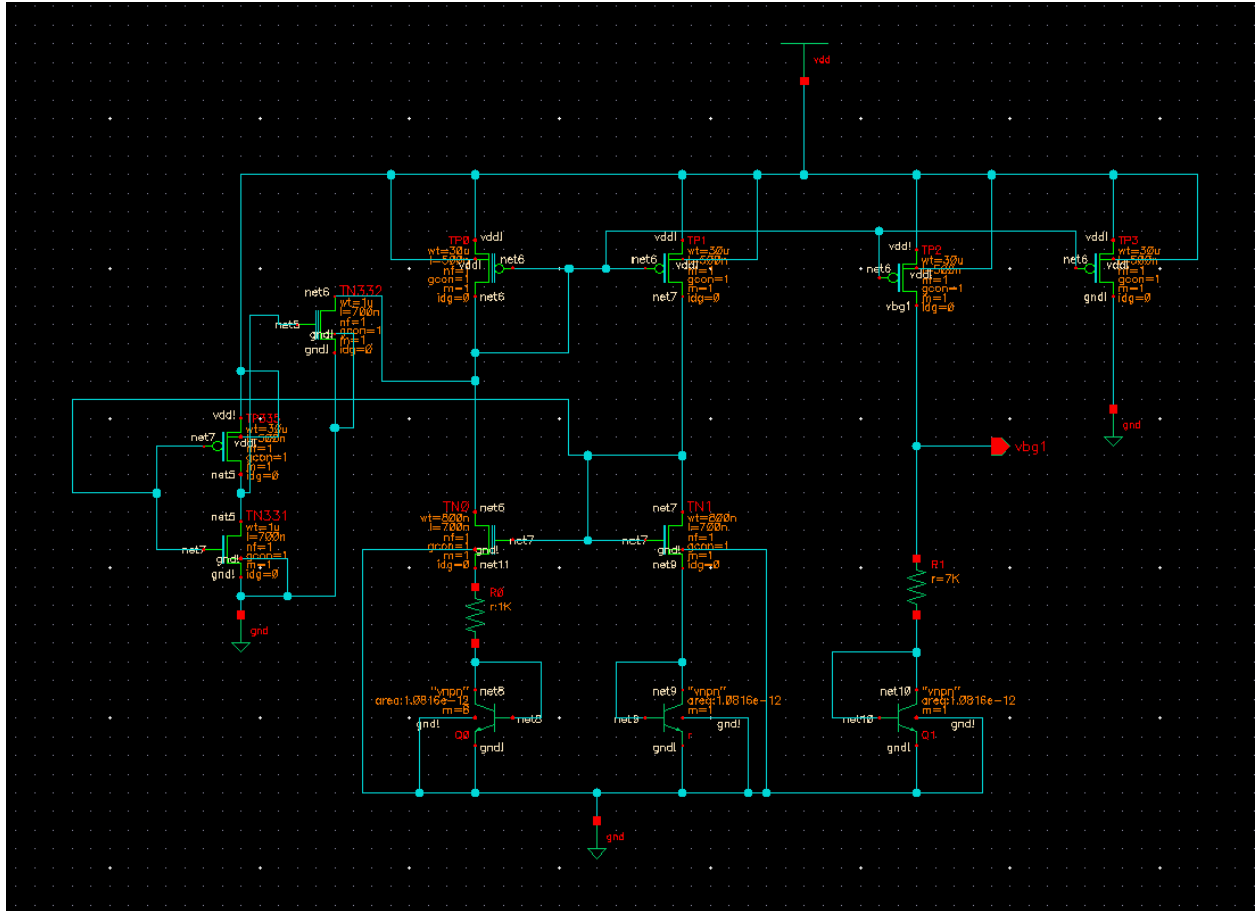
Lab Report

1. Simulate (do for both circuits), and report the following:

Complete Schematic for both the circuits:



With the start up circuit:



Specifications used for bringing 50 u current in both the arms of the branches:

TP0	PFET	30u	500n
TP1	PFET	30u	500n
TP2	PFET	30u	500n
TP3	PFET	30u	500n
TN0	NFET	800n	700n
TN1	NFET	800n	700n

Device	Type	Width (w)	Length (l)	Area
TP330	PMOS	5u	500n	-
TP331	PMOS	5u	500n	-

(NMOS)	NMOS	800n	700n	-
Q2	NPN	-	-	1.0816e-12
Q3	NPN	-	-	1.0816e-12
Circuit 1		R1	1 k Ω	
Circuit 1		R2	7 k Ω	
Circuit 2		R1	15 k Ω	
Circuit 2		R2	5 k Ω	
Circuit 2		RL	20 k Ω	

a. Output voltage vs temperature. (-40 C to +85 C - the typical consumer electronics temperature specification).

Choosing Analyses -- ADE L (1)@n01-zeus.olympus.ece.tamu.edu

Analysis

☐ tran	☒ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ lf	☐ sp
☐ envlp	☐ pss	☐ pac	☐ pstb
☐ pnoise	☐ pxf	☐ psp	☐ qpss
☐ qpac	☐ qpnoise	☐ qpxf	☐ qpssp
☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
☐ hbasp	☐ hbxf		

DC Analysis

Save DC Operating Point ☒

Hysteresis Sweep ☐

Sweep Variable

☒ Temperature

☐ Design Variable

☐ Component Parameter

☐ Model Parameter

Sweep Range

☒ Start-Stop

Start Stop

☐ Center-Span

Sweep Type



☒ Step Size

☐ Number of Steps

Add Specific Points ☐

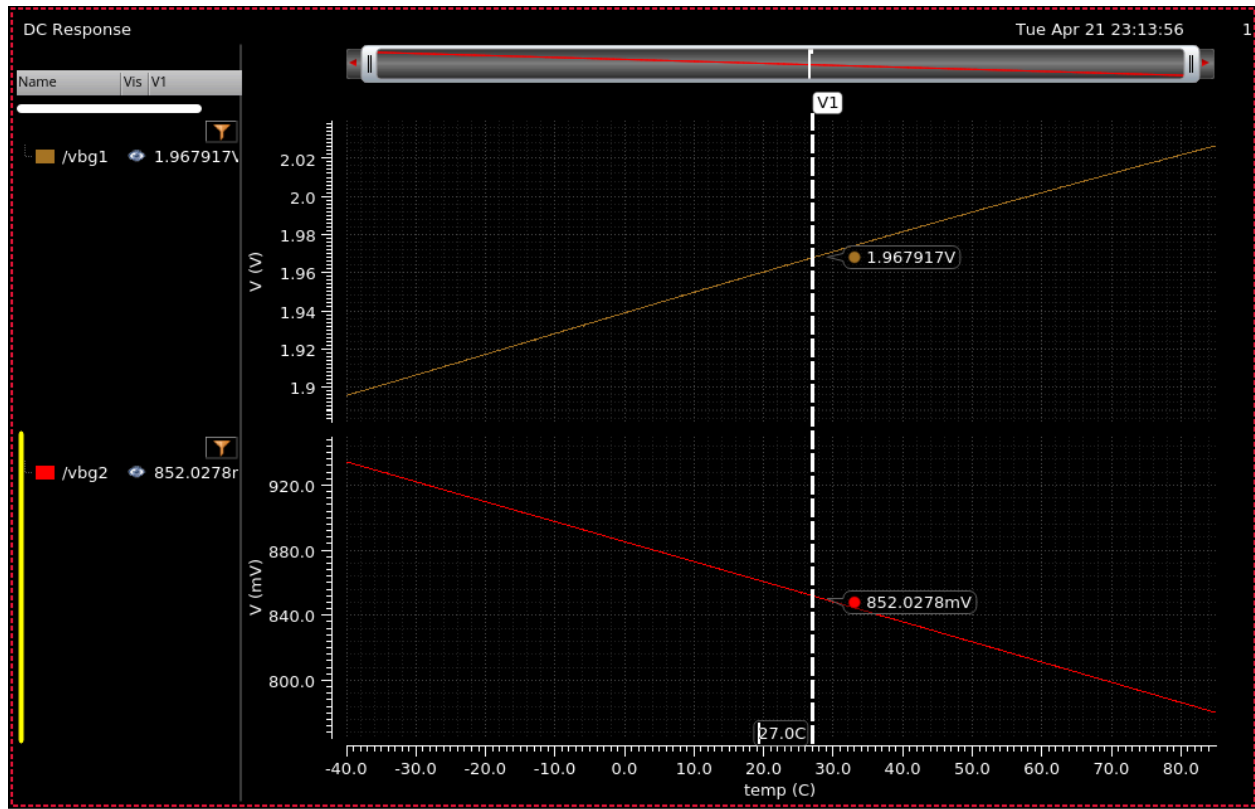
Add Points By File ☐

Enabled ☒

Options...

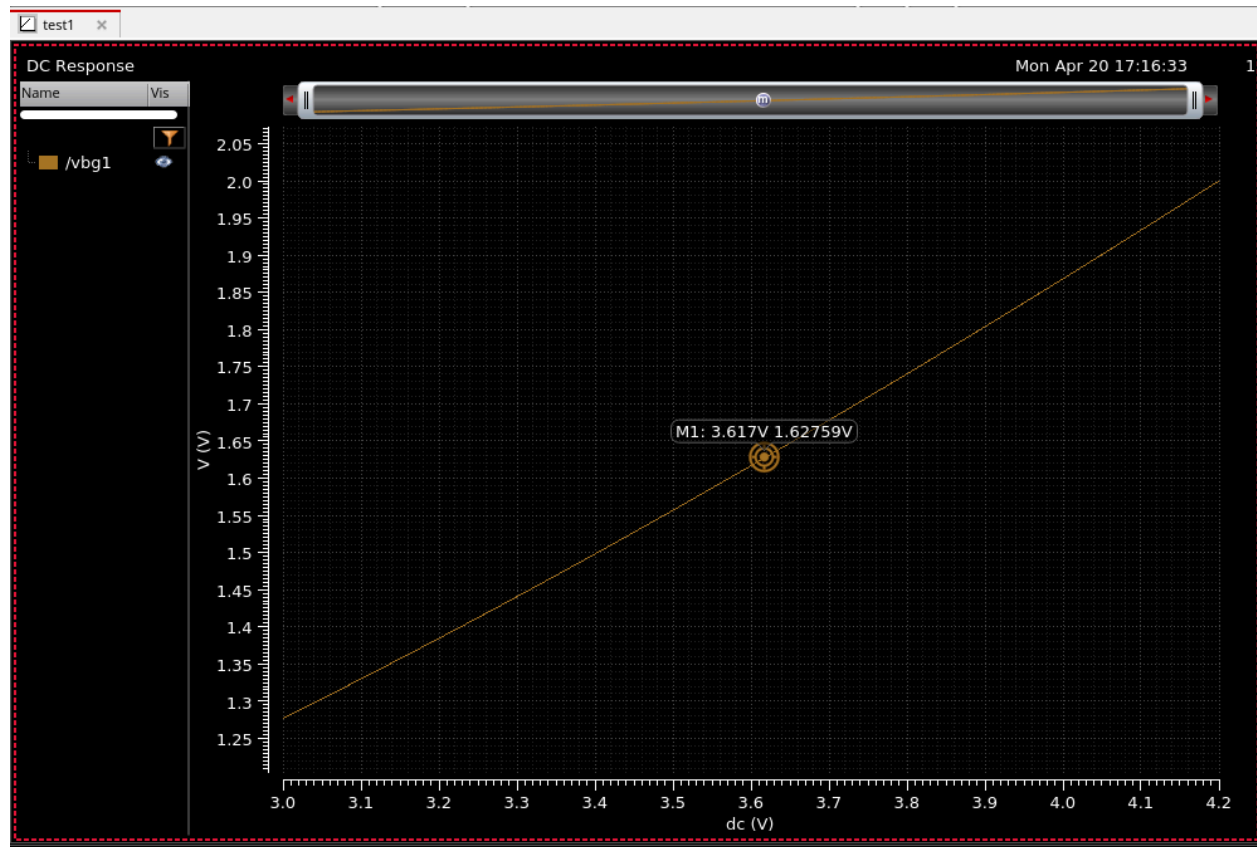
OK Cancel Defaults Apply Help

For circuit 1 and circuit 2:

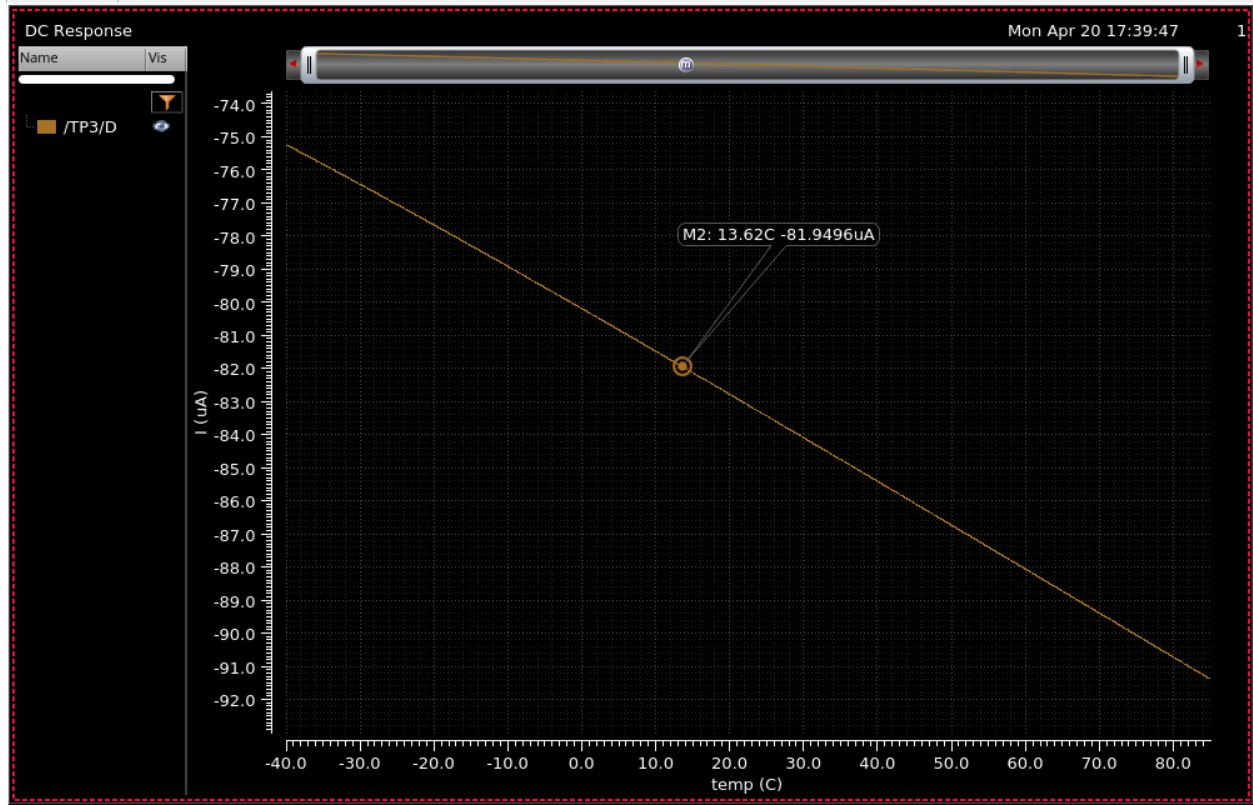


b. Output voltage vs supply voltage (3 V to 4.2 V - corresponding to a commercial Li-Ion battery)

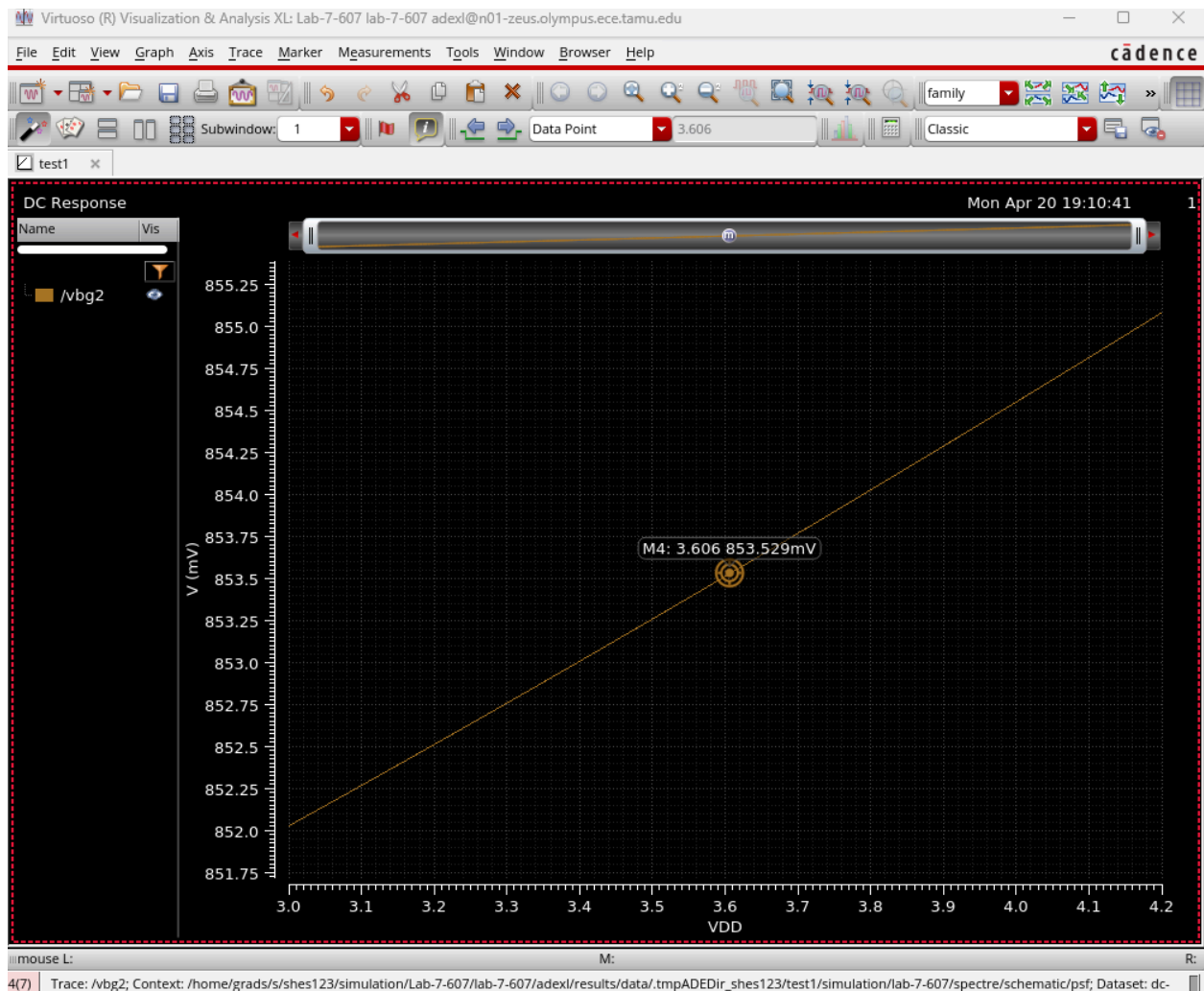
For circuit 1:



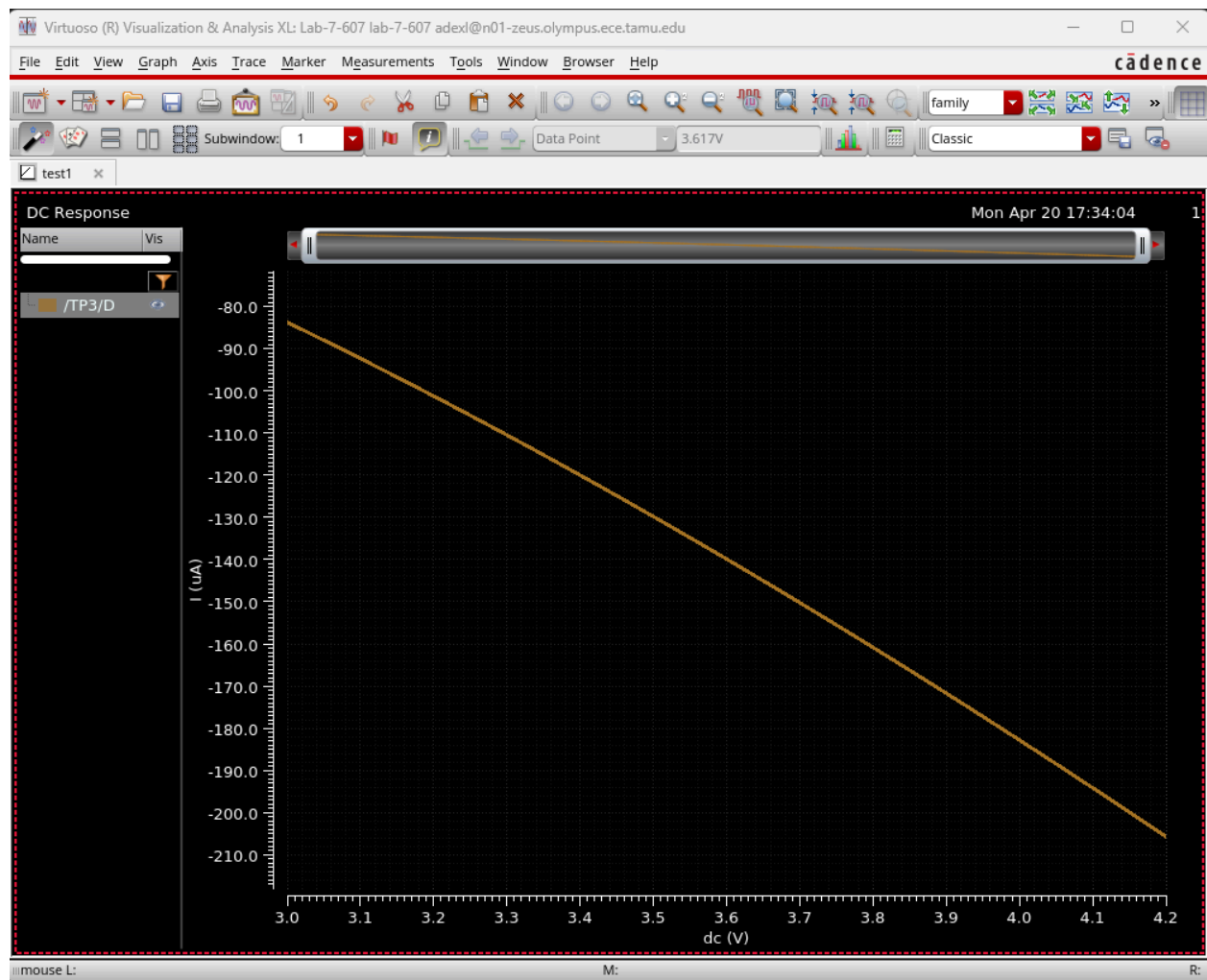
c. Reference current vs. temperature
For circuit 1:



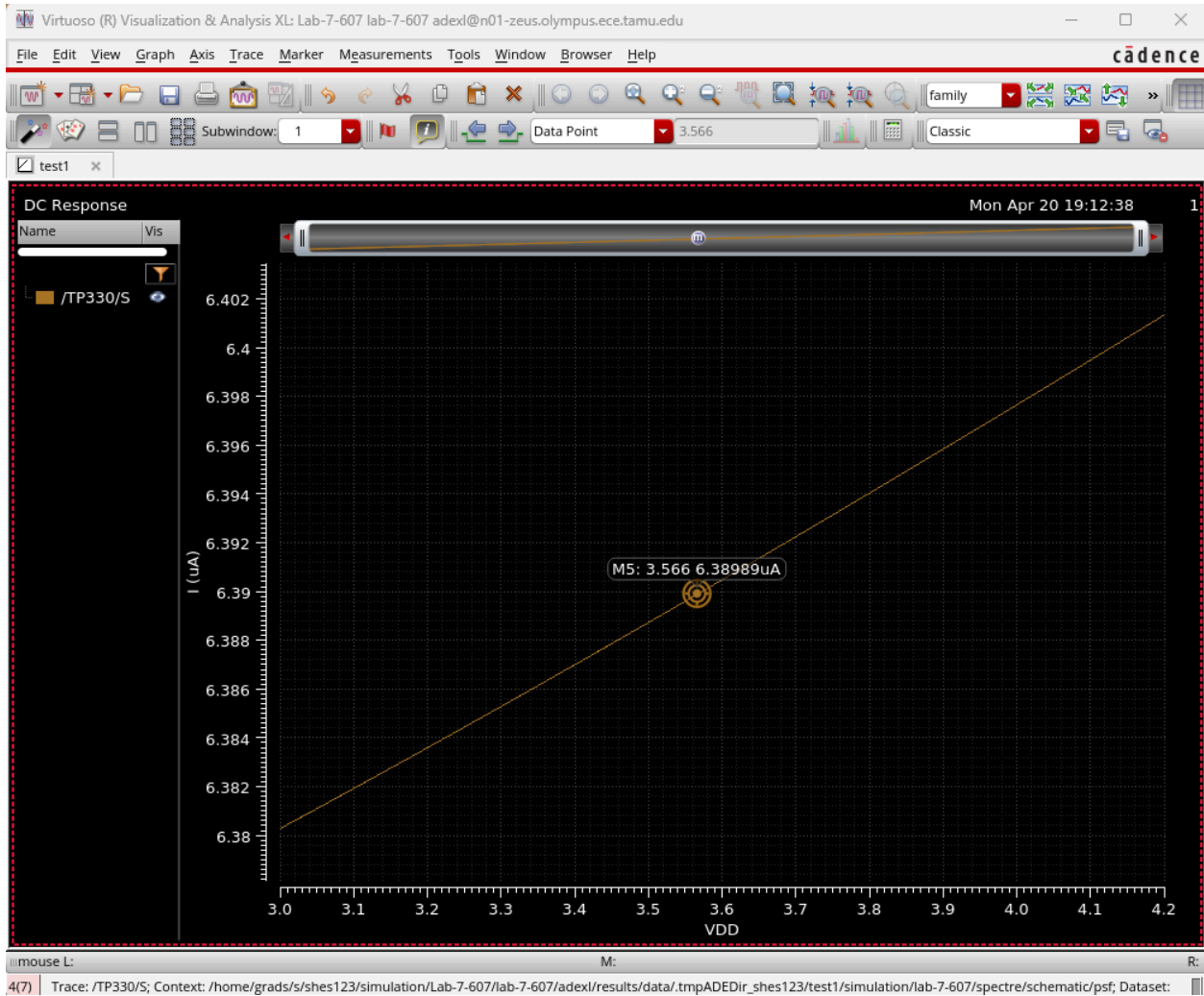
Circuit 2:

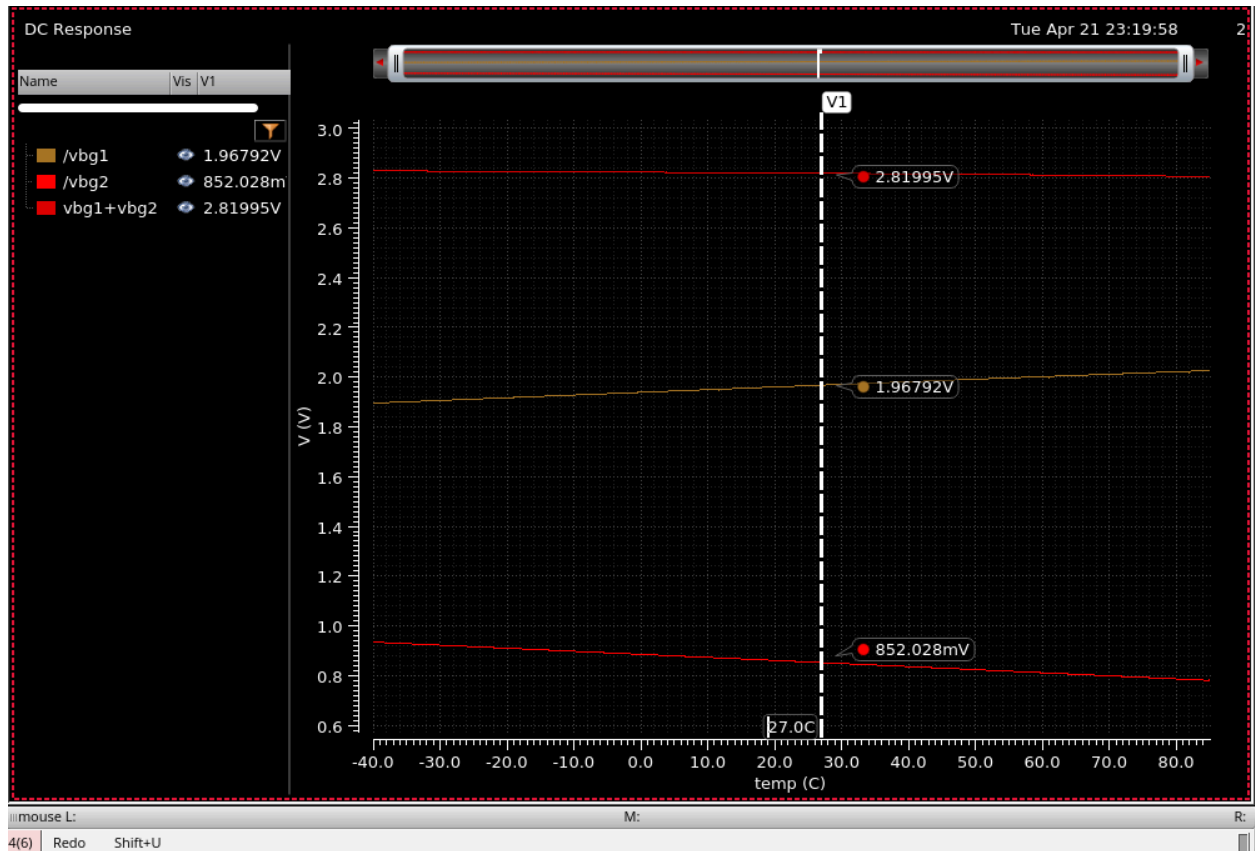


d. Reference current vs. supply variations
For circuit 1:



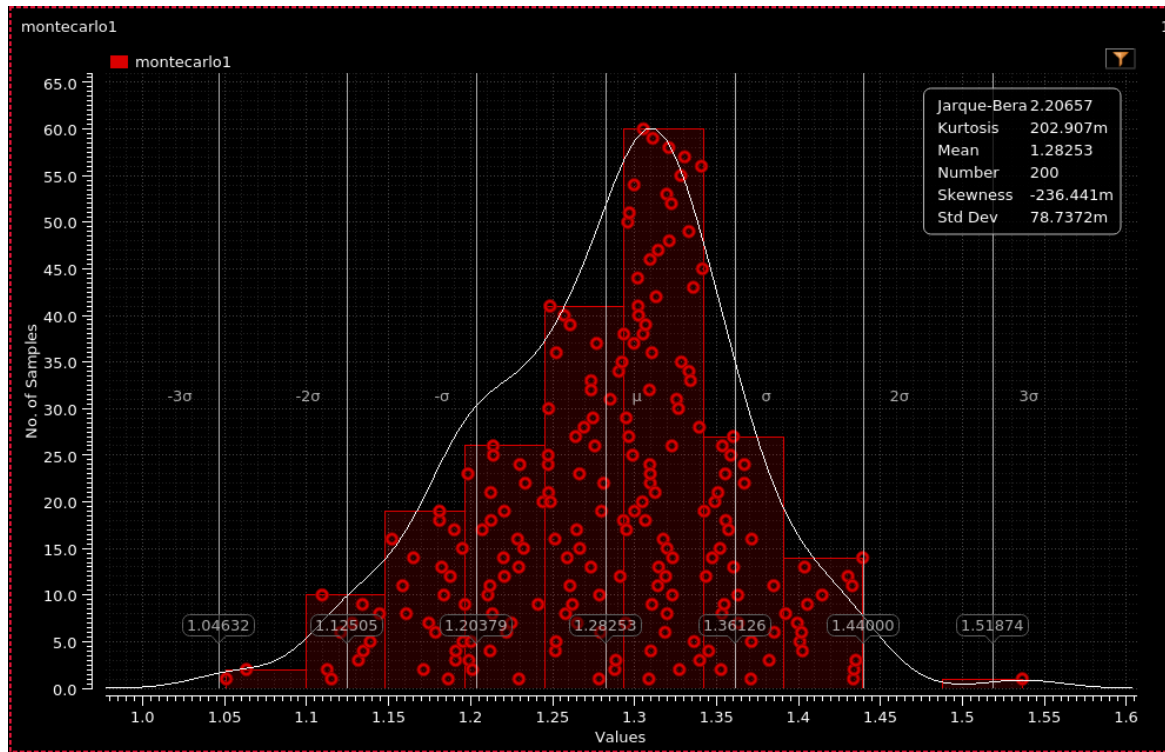
Circuit 2:



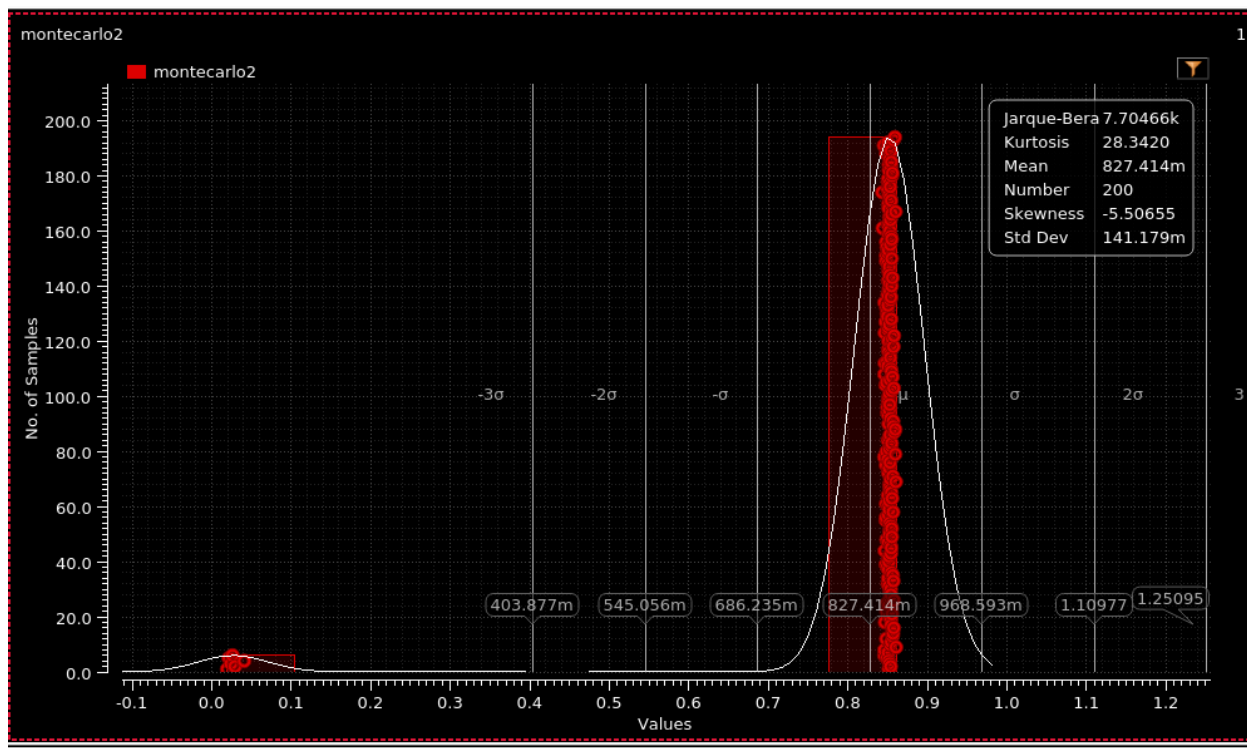


e. Monte-Carlo simulation to specify bandgap voltage accuracy (as percentage of output voltage)

For Circuit 1:



For Circuit 2:

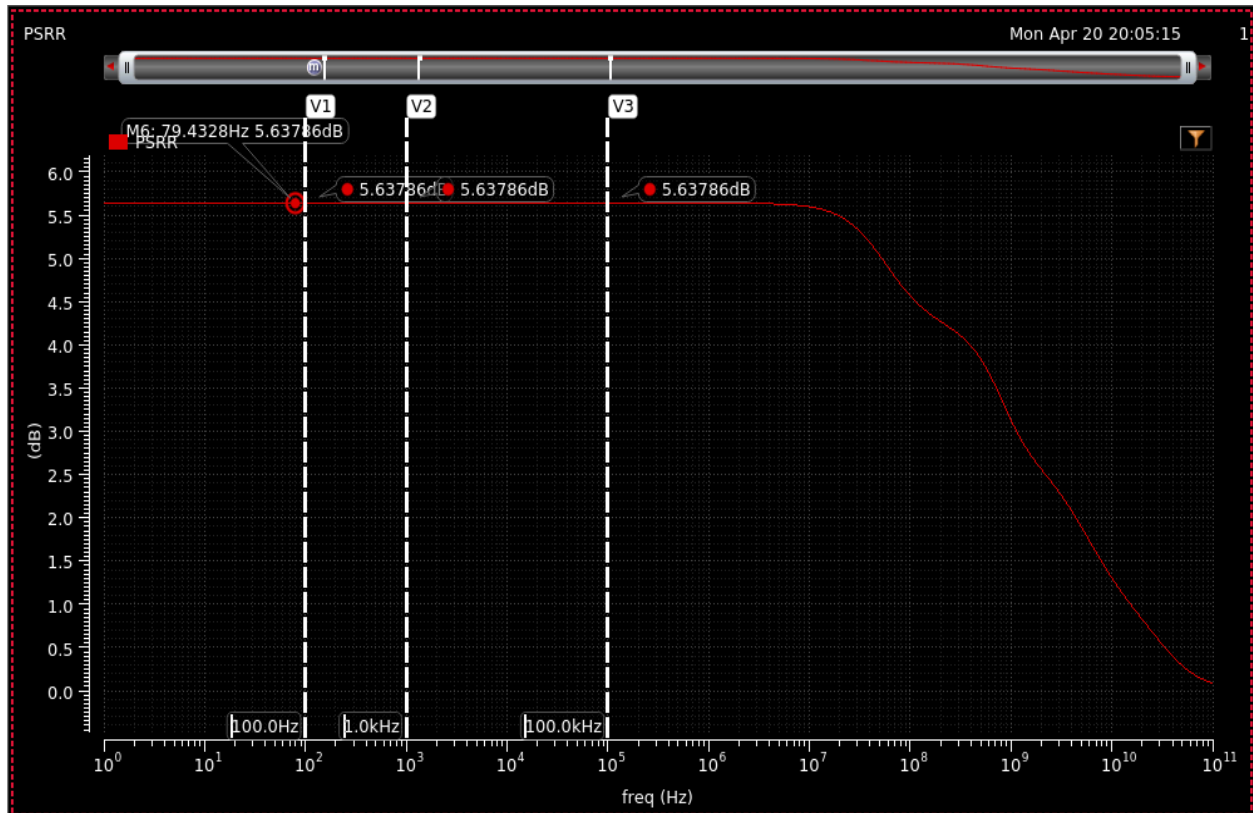


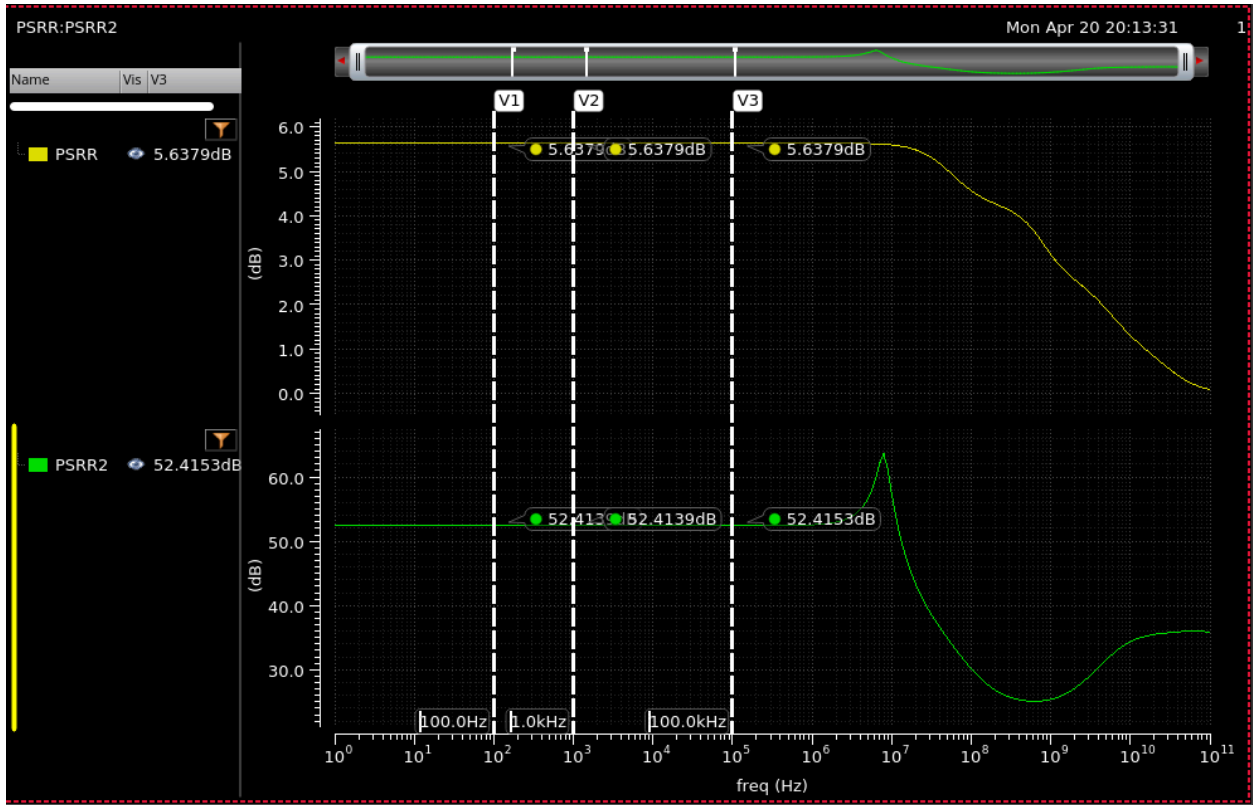
bandgap voltage accuracy:

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
- test1									
	montecarlo1	100% (200/200)	1.051	info	1.537	1.283	78.74m		0
	montecarlo2	100% (200/200)	19.36m	info	860.5m	827.4m	141.2m		0

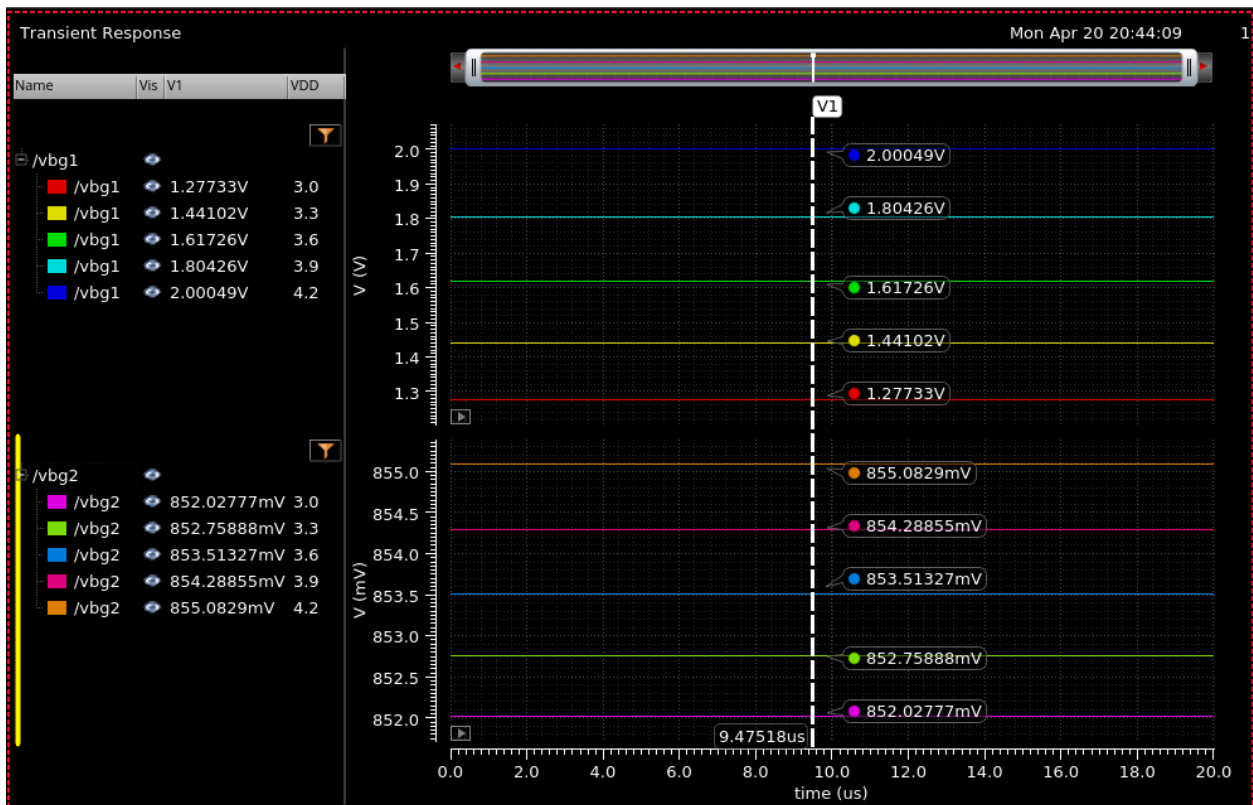
Outputs				
Name/Signal/Expr	Value	Plot	Save	Save Options
1 vbg1		☐	☐	allv
2 current		☐	☐	
3 TP3/D		☐	☐	no
4 vbg2		☐	☐	allv
5 TP330/S		☐	☒	yes
6 montecarlo1	1.277	☒	☐	
7 montecarlo2	852m	☒	☐	
8 deviation vbg1	17.7	☒	☐	
9 accuracy-vb2	997.7m	☒	☐	

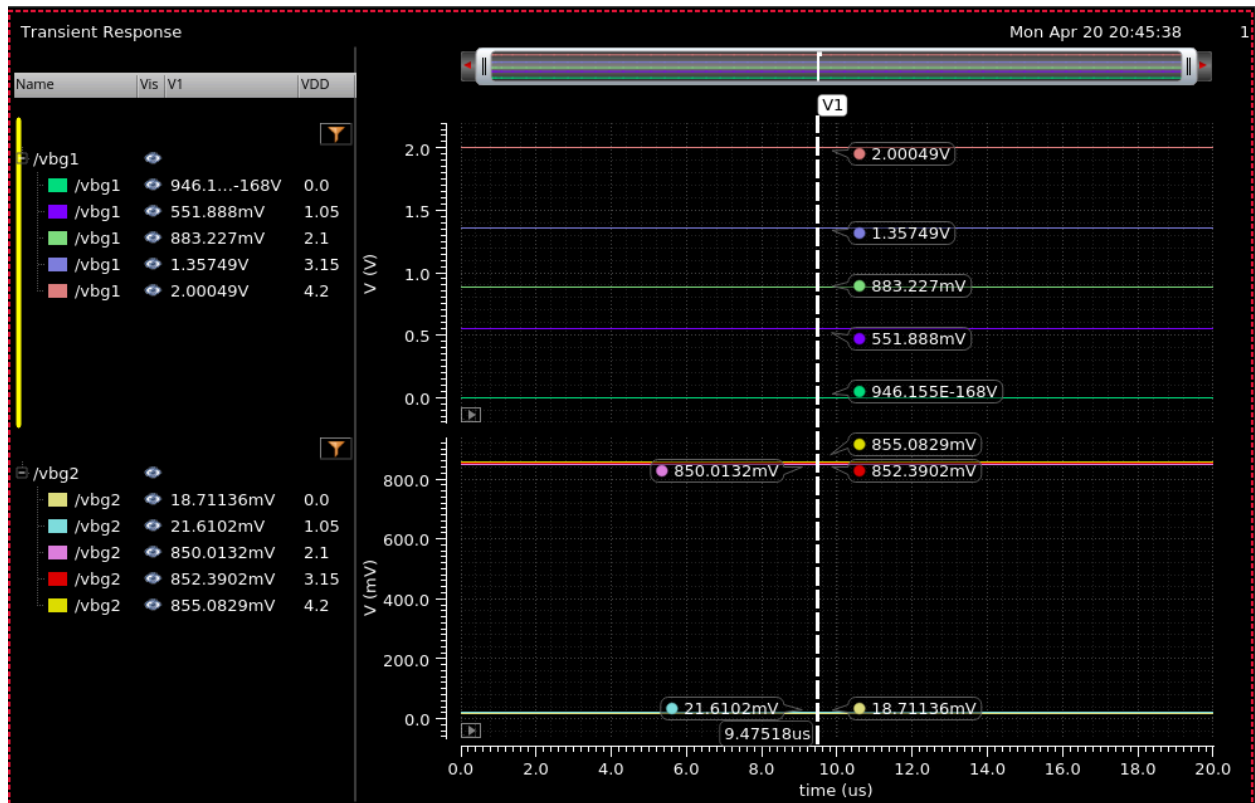
f. PSRR at 100Hz, 1kHz, and 100 kHz





g. At nominal temp, apply a ramp on Vdd (0 V to 4.2 V) and show the transient bandgap voltage (to show the working of start-up circuit)





h. Find maximum accuracy as percentage of output voltage.

Outputs				
Name/Signal/Expr	Value	Plot	Save	Save Options
1 vbg1		☐	☐	allv
2 current		☐	☐	
3 TP3/D		☐	☐	no
4 vbg2		☐	☐	allv
5 TP330/S		☐	☒	yes
6 montecarlo1	1.277	☒	☐	
7 montecarlo2	852m	☒	☐	
8 deviation vbg1	17.7	☒	☐	
9 accuracy-vb2	997.7m	☒	☐	

i. Compare both topologies (attach one table) and indicate advantages and disadvantages.

j. For circuit 2 only: Show the Loop Gain (magnitude and phase plots) and report the stability metrics

Loop gain of the second circuit:

